ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(c)** accumulation of food	Phloem blocked.
3	**(b)** glandular epithelium	Secretion.
4	**(b)** sensory neuron, relay neuron and motor neuron	Reflex arc.
5	**(a)** lignin	Lignified walls.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(b)** xylem is intact	Water still moves.
8	**(c)** a single nucleus and spindle shape	Unstriated.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Growth requires new cells, which are produced by dividing meristematic cells.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; the nature of xylem does not explain the function of phloem.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Secondary xylem accumulates every year (1); Lignified walls give strength (1).	1+1
12	Xylem upward only, phloem both directions (1); Xylem passive; phloem needs energy (1).	1+1
13	No growth in length or girth (1); No new organs would develop (1).	1+1
14	Place a twig in red ink (1); Xylem gets coloured red (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Potato turns blue-black as it stores starch (1); Carrot gives less blue-black (1); Parenchyma stores food (1).	1+1+1
16	Constant energy demand (1); Continuous contraction (1); Supplies ATP (1).	1+1+1
17	Vascular cambium: secondary xylem and phloem (1); Cork cambium: cork (1); Together form secondary growth (1).	1+1+1
	OR: Dicot stem (1); Vascular bundles in a ring (1); Open bundles with cambium (1).	1+1+1
18	Apical meristem adds height (1); Lateral meristem adds girth (1); Annual rings record growth (1).	1+1+1
	OR: Day: turgid, stoma open (1); Night: flaccid, stoma closed (1); Controls transpiration (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Intercalary meristem (1). (ii) Lateral meristem (cambium) (1). (iii) Apical meristem (1). (iv) The intercalary meristem is at the base of the leaves or internodes, below the cut, so it is not removed (1), whereas the apical meristem at the tip is removed when the tip is cut (1).	1+1+1+2
20	Dicot: ring of open bundles (1); Monocot: scattered closed bundles (1); Cambium present in dicots (1); No secondary growth in monocots (1); Examples (1).	1+1+1+1+1
	OR: Plants: meristems regenerate (1); Animals: epithelium regenerates quickly (1); Bone and liver: moderate (1); Neurons: limited (1); Stem cell therapy (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that xylem conducts food and phloem conducts water; xylem: water and minerals, phloem: food.
- Calling xylem vessels and sieve tubes both living; xylem conducting elements are dead, sieve tubes are living but without a nucleus and depend on companion cells.
- Mixing up tendons and ligaments (tendon: muscle to bone; ligament: bone to bone).

- Saying cardiac muscle is voluntary or unbranched; it is involuntary, striated and branched.
- Confusing apical meristem (length), lateral meristem (girth) and intercalary meristem (regrowth at base); also writing that meristem cells have large vacuoles (they have dense cytoplasm and small or no vacuoles).
- Listing blood as an epithelial tissue; it is a connective tissue with a fluid matrix (plasma).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(b)** vascular cambium towards the inside	Wood.
3	**(c)** fusion of many cells during development	Syncytium.
4	**(c)** synapse	Junction.
5	**(a)** starch in parenchyma	Blue-black.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(b)** outside	Protection.
8	**(d)** cardiac muscle	Contractile.
9	**(c)** A is true but R is false.	Cardiac muscle cells are branched and usually have a single nucleus, so R is false.
10	**(d)** A is false but R is true.	Blood belongs to connective tissue, not epithelial tissue, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Thick cuticle reduces water loss (1); Sunken stomata reduce transpiration (1).	1+1
12	RBCs live about 120 days (1); WBCs fight infection and die (1).	1+1
13	Sieve tubes lack a nucleus (1); Companion cells control their functions (1).	1+1
14	Bone has a hard matrix and less blood supply (1); Needs new bone cells to deposit minerals (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Skeletal: many nuclei, striated (1); Smooth: single nucleus, unstriated (1); Cardiac: single or two nuclei, striated, branched (1).	1+1+1
16	Chlorenchyma (parenchyma with chloroplasts) (1); Photosynthesis (1); Found in leaves and green stems (1).	1+1+1
17	Jute and flax from the stem sclerenchyma (1); Cotton from the seed hairs (1); Used for fabrics (1).	1+1+1
	OR: Sugary sap in the phloem lies in the bark (1); Xylem water is deeper (1); The cut opens the phloem (1).	1+1+1
18	Stratified squamous epithelium protects (1); Nerve endings sense (1); Sweat glands and blood vessels regulate temperature (1).	1+1+1
	OR: Provides strength (1); Lignin makes it hard (1); Timber is mostly secondary xylem and fibres (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Xylem (1). (ii) Tracheids and vessels (1). (iii) Upward, from roots to leaves (1). (iv) Phloem carries food prepared in the leaves, not water from the soil, so it takes up no ink (1); it transports food from leaves to other parts through sieve tubes assisted by companion cells (1).	1+1+1+2
20	Sensory: receptor to spinal cord (1); Relay in spinal cord (1); Motor: to muscle (1); Muscle contracts (1); Reflex is fast and automatic (1).	1+1+1+1+1
	OR: Select explant (1); Sterile nutrient medium with hormones (1); Callus forms (1); Plantlets grown and transferred (1); Applications: orchids, bananas, conservation (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that xylem conducts food and phloem conducts water; xylem: water and minerals, phloem: food.
- Calling xylem vessels and sieve tubes both living; xylem conducting elements are dead, sieve tubes are living but without a nucleus and depend on companion cells.

- Mixing up tendons and ligaments (tendon: muscle to bone; ligament: bone to bone).
- Saying cardiac muscle is voluntary or unbranched; it is involuntary, striated and branched.
- Confusing apical meristem (length), lateral meristem (girth) and intercalary meristem (regrowth at base); also writing that meristem cells have large vacuoles (they have dense cytoplasm and small or no vacuoles).
- Listing blood as an epithelial tissue; it is a connective tissue with a fluid matrix (plasma).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** A is true but R is false.	A is true; R is false: cardiac muscles are involuntary.
2	**(b)** its age	One ring per year.
3	**(b)** transparent epithelium and connective tissue	Transparent.
4	**(d)** electrical and chemical changes	Impulse.
5	**(d)** xylem carries water	Transport.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(c)** favourable growth conditions	Good season.
8	**(d)** sweep dust and mucus	Cleaning.
9	**(a)** Both A and R are true and R is the correct explanation of A.	When guard cells swell the pore opens, and when they lose water it closes.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but they describe different structures.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Oxygen and glucose for energy (1); Remove carbon dioxide and wastes (1).	1+1
12	Muscles need much energy for contraction (1); Mitochondria release the energy (1).	1+1
13	Nervous: conduction of impulses (1); Muscular: contraction and movement (1).	1+1
14	Reduces friction (1); Absorbs shock (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Epidermis admits light (1); Palisade tissue photosynthesises (1); Xylem brings water, phloem takes away sugars (1).	1+1+1
16	Immune system recognises foreign tissue (1); Rejection can occur (1); Matching reduces rejection (1).	1+1+1
17	Nervous tissue (1); Impulses cannot pass to muscles (1); Paralysis and loss of sensation (1).	1+1+1
	OR: Biceps contracts, triceps relaxes (1); Tendons pull the forearm bone (1); Hinge joint with ligaments and cartilage (1).	1+1+1
18	Can differentiate into many cell types (1); Replace damaged cells (1); Used in bone marrow transplant (1).	1+1+1
	OR: Cell: gland cell (1); Tissue: epithelium (1); Organ: stomach; system: digestive (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Cardiac muscle (1). (ii) Skeletal (striated) muscle (1). (iii) Smooth (unstriated) muscle (1). (iv) It works without conscious control, so it is involuntary (1); its cells are striated, branched and joined at intercalated discs so that the heart contracts together rhythmically (1).	1+1+1+2
20	Root: epidermis with root hairs (1); Stem: epidermis with cuticle (1); Vascular bundles: alternate vs same radii (1); Pith small in roots, large in stems (1); Functions (1).	1+1+1+1+1
	OR: Skin: stratified squamous (1); Intestine: columnar with microvilli (1); Lung: thin squamous (1); Blood vessels: squamous endothelium (1); Structure fits function (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that xylem conducts food and phloem conducts water; xylem: water and minerals, phloem: food.
- Calling xylem vessels and sieve tubes both living; xylem conducting elements are dead, sieve tubes are living but without a nucleus and depend on companion cells.
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- Confusing apical meristem (length), lateral meristem (girth) and intercalary meristem (regrowth at base); also writing that meristem cells have large vacuoles (they have dense cytoplasm and small or no vacuoles).
- Listing blood as an epithelial tissue; it is a connective tissue with a fluid matrix (plasma).

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